VISHWAKARMA UNIVERSITY, PUNE • DEPARTMENT OF COMPUTER ENGINEERING • A.Y. 2026-2027 • DIVISION B (GROUP 21)

MINOR PROJECT PRESENTATION POSTER

MINDGUARD AI: SENTIMENTAL ANALYSIS FOR EARLY MENTAL HEALTH DETECTION

A Deep Bidirectional Transformer (BERT) Architecture for 7-Class Affective Classification, Explainable Attention Saliency & Automated Crisis Triage

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Live System: 31232258arnav.github.io/MindGuard_AI/
Repository: github.com/31232258arnav/MindGuard_AI

Evaluation Benchmark: 83.00% Test Accuracy • 0.8299 Weighted F1 (10,536 Samples)

1. PROBLEM STATEMENT & CLINICAL RATIONALE

Clinical Reality: Over 1 billion individuals globally suffer from mental health disorders (WHO). Acute shortages of clinical psychiatrists, societal stigma, and prohibitive consultation costs prevent over 75% from seeking timely diagnosis. Suicide is the 3rd leading cause of death among youth aged 15–29.
The NLP Challenge: Early psychological distress hides in *informal, unstructured digital text* (journals, messaging, peer posts). Conventional sentiment tools use simplistic binary labels (positive/negative), failing to detect the subtle co-morbid affective states characteristic of clinical psychiatric distress.

2. CORE RESEARCH & ENGINEERING OBJECTIVES

- **7-Class Deep Classification:** Fine-tune BERT-base on 7 mutually exclusive affective categories.
- **Explainable AI (XAI):** Extract self-attention token weights for clinical explainability heatmaps.
- **Validated Psychometrics:** Digitize standard DSM-5 clinical batteries (PHQ-9 and GAD-7).
- **Empathetic Conversational Agent:** Develop *MindBot* with strict clinical domain boundary.
- **Instant Emergency Triage:** Hardcoded helpline escalation (Tele-MANAS: 14416) for suicidal distress.
- **Zero-Cost Cloud Deployment:** Resilient tri-tier topology (GitHub Pages, Cloudflare Tunnel, Render).

3. DATASET CORPUS & STRATIFIED DISTRIBUTION

Trained on **52,680 curated clinical statements** with stratified 72% Train (37,930), 8% Val (4,214), and 20% Test:

Affective Category	Support	Share	Clinical Risk Level
Normal	3,269	31.0%	Safe / Baseline
Depression	3,081	29.2%	Elevated Clinical Strain
Suicidal	2,130	20.2%	**CRITICAL EMERGENCY**
Anxiety	768	7.3%	Mild Distress Monitoring
Bipolar	555	5.3%	Elevated Clinical Strain
Stress	518	4.9%	Situational Strain
Personality disorder	215	2.0%	Regulation Needed
Total Held-Out Test	**10,536**	**100%**	Full Stratified 72/8/20 Split

4. DEEP TRANSFORMER NEURAL ARCHITECTURE

Neural Backbone: BERT-base-uncased (12 layers, 768 hidden, 12 attention heads, 110M parameters).
Input Representation: [CLS] token embedding passed to dropout (p=0.3) & Linear(768 → 7) with Softmax.
Optimization Protocol: AdamW (ε = 1e-8), Linear Warmup & Decay (lr = 2e-5), Max Seq = 128 tokens, Batch = 16.
Checkpoint Selection: Trained across 7 epochs (~16,597 gradient steps); best weights saved at **Epoch 5** (Val Acc: 82.80%).

5. END-TO-END DISTRIBUTED SYSTEM TOPOLOGY

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| 1. CLIENT TIER (SPA on GitHub Pages Edge) |
| - Single Page App (HTML5 / Tailwind CSS / ES6+ JS / Web Speech API) |
| - Interactive Token Saliency Canvas • PHQ-9 / GAD-7 • Automated SOAP |
| - Client-Side Fallback Engine (Zero-dependency embedded JS classifier) |
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■ ■ (Offline)
▼ (HTTPS / TLS 1.3) ▼
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| 2. EDGE & TUNNEL GATEWAY TIER | | CLIENT-SIDE HEURISTIC ENGINE |
| - Cloudflare Secure Edge Tunnel | | - Zero-latency local regex |
| - Dynamic endpoint autodiscovery | | - Operates without backend |
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▼ (FastAPI ASGI)
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| 3. ASYNCHRONOUS APPLICATION TIER (Uvicorn) |
| - REST Endpoints: /api/analyze • /api/chat • /api/batch • /api/benchmarks|
| - Document Parser: Multi-format PDF, DOCX, CSV, TXT audit processor |
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| 4. BERT TRANSFORMER | | 5. GEMINI LLM | | 6. CLINICAL SAFETY CORE |
| - PyTorch GPU Engine | | - Empathetic | | - Tele-MANAS (14416) |
| - 7-Class Softmax Head | | MindBot Brain | | - Suicide Crisis Triage |
| - Saliency Attention | | - Coping Guide | | - Automated SOAP Notes |
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6. CLINICAL PSYCHOMETRIC BATTERIES (PHQ-9 & GAD-7)

PHQ-9 (Patient Health Questionnaire-9): Standardized clinical instrument for depression severity scoring (0–27). Evaluates anhedonia, sleep disturbances, fatigue, appetite, self-worth, concentration, psychomotor slowing, and suicidal ideation.
GAD-7 (Generalized Anxiety Disorder-7): Standardized instrument for anxiety measurement (0–21). Quantifies uncontrollable worry, restlessness, irritability, and somatic panic symptoms.
Automated DSM-5 Stratification: Scores automatically categorize severity into Minimal, Mild, Moderate, or Severe, generating structured clinical recommendations.

7. MINDBOT NEURAL CHAT & CRISIS HELPLINE TRIAGE

Empathetic Persona Guardrails: *MindBot* integrates multi-provider LLMs (Google Gemini 2.5 Flash / Claude 3.5 Sonnet) strictly bounded to mental health dialogue. Rejects non-clinical queries (coding, trivia) and avoids clinical diagnostic claims.
Evidence-Based Grounding: Recommends 4-7-8 vagal nerve breathing, 5-4-3-2-1 sensory grounding, and behavioral micro-activation.
Hardcoded Emergency Suicide Triage: When acute suicidal ideation is detected, probabilistic small talk is bypassed. A high-priority emergency banner immediately surfaces verified 24/7 national mental health helplines:
- **Tele-MANAS (Govt. of India):** 14416 / 1800-891-4416 (Toll-Free, 24/7, 20+ Languages)
- **KIRAN (Ministry of Social Justice):** 1800-599-0019 (24/7 National Psychological Helpline)
- **Vandrevala Foundation Helpline:** +91 9999 666 555 (24/7 Crisis Support)

8. MODEL EVALUATION & BENCHMARK METRICS

Evaluation on **10,536 held-out test statements** unseen during training:

Target Class	Precision	Recall	F1-Score	Support	Clinical Risk Level
Normal	0.9576	0.9523	**0.9549**	3,269	Safe Baseline
Anxiety	0.8753	0.8958	**0.8855**	768	Mild Distress
Bipolar	0.8330	0.8721	**0.8521**	555	Elevated Strain
Depression	0.7712	0.7790	**0.7751**	3,081	Elevated Strain
Personality	0.8093	0.7302	**0.7677**	215	Regulation Needed
Stress	0.7866	0.7471	**0.7663**	518	Situational Strain
Suicidal	0.7151	0.7117	**0.7134**	2,130	**CRITICAL CRISIS**
Overall Accuracy	—	—	**83.00%**	**10,536**	Macro F1: **0.8164**
Weighted F1-Score	**0.8301**	**0.8300**	**0.8299**	**10,536**	Latency: **~142 ms**

Key ML Findings: Top-2 differential diagnosis accuracy reaches 96.8%. High-confidence sub-cohort (softmax > 0.85) reaches 98.2% precision.

9. EXPLAINABLE AI (XAI) & CLOUD ENGINEERING

Token Attention Saliency (XAI): Softmax self-attention weights extracted from BERT's final transformer layer are projected directly onto UI tokens. Words indicating severe despair (*'unbearable'*, *'hopeless'*) highlight red, while stabilizing tokens (*'peaceful'*, *'grateful'*) highlight green, providing clinicians with instant visual audit trails.
Zero-Cost Cloud Deployment Rationale:
- **Frontend (GitHub Pages):** 100% uptime, global CDN caching, zero cold start ($0/mo).
- **GPU Bridge (Cloudflare Tunnel):** Direct SSL tunnel to local GPU instance with zero port forwarding ($0/mo).
- **Cloud API (Render & Vercel):** Containerized FastAPI ASGI backup service via render.yaml ($0/mo).
- **Offline Resilience:** Embedded JS heuristic classifier guarantees 100% operational continuity.

10. SOCIETAL IMPACT & ACADEMIC CONCLUSION

Societal Impact: MindGuard AI democratizes early mental health screening across collegiate campuses, workplace portals, and digital health clinics. By providing confidential, non-stigmatizing, 24/7 automated assessments coupled with instantaneous crisis escalation, the platform helps prevent suicide attempts and bridges the critical gap before professional psychiatric intervention.
Project Conclusion: Demonstrates that high-accuracy affective computing (83.00% accuracy, 0.8299 F1) can be achieved and deployed globally at **$0 infrastructure cost** using bidirectional transformers, explainable attention mechanics, and distributed cloud edge routing.
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